Technical Design — Ingestion Module

Theme & Epic Generation · Ingestion

1. Purpose & scope

The ingestion module converts **historical IDMT Engagement Request tickets** into a trusted corpus that the data-science (generation) module retrieves from. For every eligible ticket it produces:

1. a **condensed, durable record** of the ticket's content (system of record), and
2. each linked **Theme's** title, description, and **Value Stream** — the Business Architect's recorded Value Stream for that ticket, used as retrieval precedent.

Ingestion is **offline/batch** and contains no runtime generation logic. It writes to **Cosmos** (the system of record) and **idp_teg_data** (the retrieval index). The governed Value Stream / Stage / L2 / L3 catalogue is **not** produced here — it is the org's gold data in the **Azure SQL DB**, consumed as-is.

2. Flow

```
STAGE 0 — Ticket identification (Neo4j 5-filter funnel)
  Neo4j JIRA graph  →  one Cypher query (L2→L6 funnel)  →  list of usable IDMT ticket keys

STAGE 1 — Per-ticket ingestion (run once per identified key)
  IDMT Engagement Request (Jira)
    → fetch ER + linked Themes
    → attachment extraction (no idea-card detection)
    → raw text assembly (description + attachments → ~24k-token budget)
    → Condense (LLM)              → business-context fields + rawText
    → Theme extraction            → title, description, Value Stream (from Jira)
    → write Cosmos (ER doc + Theme docs) + upsert idp_teg_data (searchText + content_vector)
```

3. Ticket identification (which tickets we ingest)

A **distinct first stage, upstream of the per-ticket pipeline.** We do **not** ingest every IDMT ticket — we first identify the **usable Value-Stream cohort** and produce a list of their keys. The per-ticket pipeline (Stage 1) then runs once per identified key.

Identification runs against the **Neo4j JIRA graph** (env: `NEO4J_URI / USER / PASSWORD / DATABASE`) as a **single Cypher query** implementing a 5-filter funnel (L2→L6). A ticket survives to the cohort only if **all** hold:

filter	rule
L2 — is an IDMT Engagement Request, recent	`key` starts with `IDMT-`, `issueType = "Engagement Request"`, `creationDateEpoch ≥ since` (default **2023-01-01**)
L3 — not in a dead status	`status NOT IN {Cancelled, Blocked, New Request}`
L4 — is implemented by a linked issue	the IDMT has ≥1 **inbound "implemented by"** link — i.e. a Theme *implements* it. (From the IDMT's side this relationship is inward; the same link is outward "implements" on the Theme. We read it from the IDMT, so it appears inward.)
L5 — the linked issue is a live Theme	the linked key resolves to a JIRA node with `issueType = "Theme"` **and** its status is not `Cancelled`
L6 — the Theme carries a Value Stream	the Theme's `businessValueStreams` matches `…{VSR\d+}` (a valid Value Stream id is present)

The query returns the **distinct ER keys** that pass all five; that key set is the cohort the batch ingestion run consumes.

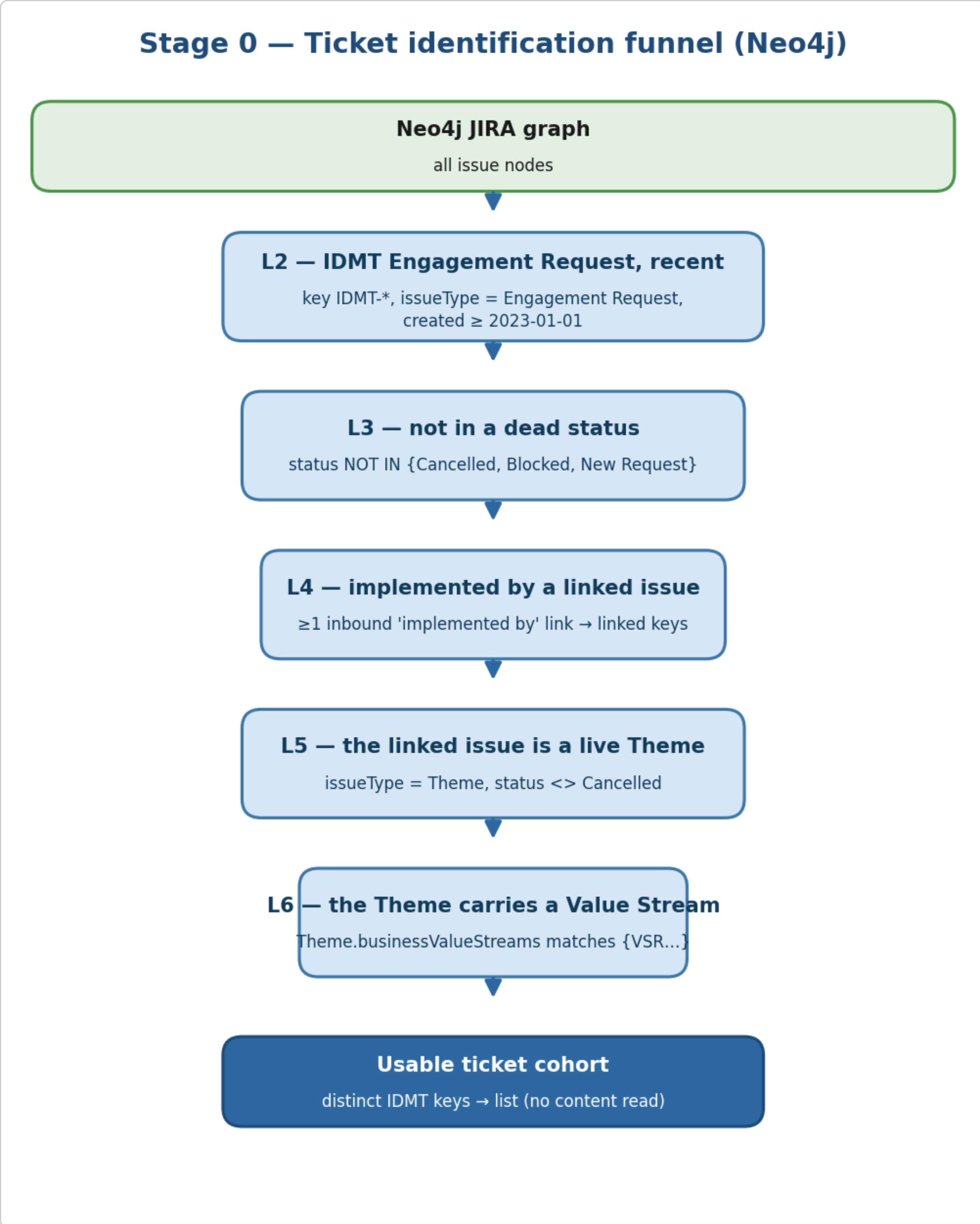
> **Status note.** The identification status exclusion is **{Cancelled, Blocked, New Request}** — the funnel rejects whole tickets that were dropped, on hold, or never started.

How the funnel filters (single pass)

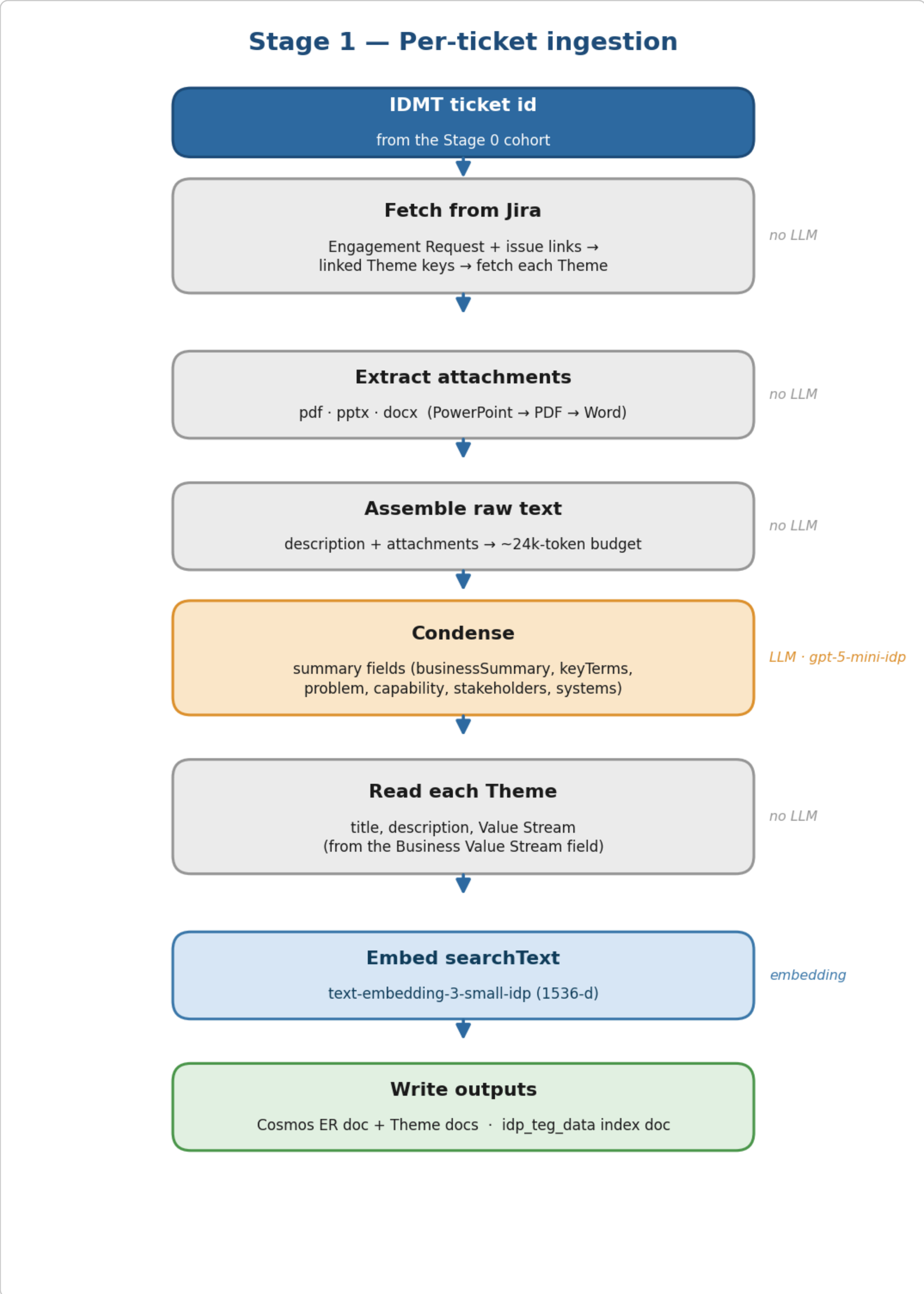
The funnel is one Cypher query — no per-ticket round trips — and each level narrows the previous set:

1. **Start from IDMT Engagement Requests (L2+L3).** Match graph nodes whose key begins `IDMT-`, whose issue type is *Engagement Request*, created on/after the cutoff, and whose status is not one of the dead states. This is the candidate pool.
2. **Pull each candidate's linked keys (L4).** From the IDMT node's inbound-link metadata, keep only the links of type *"implemented by"* and extract the linked issue keys. A candidate with no such link is dropped here — it has no implementing artifact, so no ground truth.
3. **Resolve the links to live Themes (L5).** Look up each linked key as a graph node and keep it only if its issue type is *Theme* **and** its status is not *Cancelled* — a dropped Theme is not a real label. (An "implemented by" link to a non-Theme issue does not qualify.)
4. **Require a Value Stream on the Theme (L6).** Keep the candidate only if at least one of its Themes has a `businessValueStreams` value matching `{VSR…}` — i.e. the Theme actually carries a Value Stream. A Theme with a blank or malformed Value Stream does not qualify the ticket.
5. **Return distinct, ordered IDMT keys.** A ticket that survives all four narrowings is emitted once, regardless of how many qualifying Themes it has.

The output is purely the **list of ticket keys** — no content is read in Stage 0. Reading content, attachments, and the Theme details happens per-ticket in Stage 1.


Stage 0 — Ticket identification funnel (Neo4j)
Neo4j JIRA graph
all issue nodes
L2 — IDMT Engagement Request, recent
key IDMT-*, issueType = Engagement Request, created ≥ 2023-01-01
L3 — not in a dead status
status NOT IN {Cancelled, Blocked, New Request}
L4 — implemented by a linked issue
≥1 inbound 'implemented by' link → linked keys
L5 — the linked issue is a live Theme
issueType = Theme, status <> Cancelled
L6 — the Theme carries a Value Stream
Theme.businessValueStreams matches {VSR...}
Usable ticket cohort
distinct IDMT keys → list (no content read)

4. Stage 1 — how a ticket is processed



For each key in the cohort, the per-ticket pipeline runs once. It takes **one ticket id** and assumes Stage 0 already qualified it. The steps:

1. **Fetch from Jira.** Pull the Engagement Request (its fields + issue links). The links give the **linked Theme keys**; each Theme is then fetched with its Business Value Stream field. A linked issue that is not a real Theme (no Value Stream) is dropped, so only genuine Themes remain.
2. **Extract attachments** and **assemble the raw text** (§4.1, §4.2).
3. **Condense** the raw text into the business-context fields (§4.3).
4. **Read each Theme's** title, description, and Value Stream (§4.4).
5. **Write** the Cosmos Engagement-Request document, one Cosmos Theme document per linked Theme, and the AI-search index document (embedded) (§5).

The sub-steps below detail each part.

4.1 Attachment extraction

The business content comes from the ticket's **attachments**, fetched directly — there is **no idea-card detection**; every supported attachment is extracted.

- **Supported formats & extraction libraries:**

format	library
`.pdf`	**pypdfium2** (PDFium via ctypes — fast, releases the GIL while parsing)
`.pptx`	**python-pptx**
`.docx`	**python-docx**

- **Priority order: PowerPoint → PDF → Word.** Attachments are ordered by format priority (`.pptx` first, then `.pdf`, then `.docx`); within the same format, Jira's original order is kept. PowerPoint is highest because SMEs confirmed it is the most common idea-card format. This order is what the raw-text packing (§4.2) follows when the budget is tight.

- Legacy binary `.ppt` / `.doc` and image-only files would yield no text — but the EDA found **none** of these in the corpus, so all attachments in scope are text-extractable. No OCR.

4.2 Raw text assembly

The Jira **description** and the extracted attachment text are concatenated into a single **raw text** blob, greedily packed into a **~24k-token budget**, attachments taken in the priority order from §4.1 (PowerPoint → PDF → Word). The description is part of that budget — there is no separate budget for it. Highest-priority content is never displaced or truncated to fit a lower-priority attachment; the token budget is the only cap.

4.3 Condense (LLM)

A single LLM pass extracts structured business context from the raw text, so downstream steps don't re-process it. It produces the LLM-derived fields stored on the IDMT document (§4.4), and the raw text is carried through as `rawText`.

- **Model:** `gpt-5-mini-idp` (the condense/summarization model).

- **Output** is a typed structured-output schema, not free-form JSON in the prompt.

4.4 Fields extracted

We extract content directly from Jira (and the LLM condense pass). We do **not** extract or store Epics, Stages, L2/L3 capabilities, or Business Needs — only the Theme's title, description, and Value Stream.

IDMT Engagement Request — from Jira

field	source
`key`	Jira issue key (IDMT-####)
`sourceId`	stable Jira internal id
`description`	Jira description
`summary`	ticket title
`creationDate`	source creation date
`insightsTime`	source last-updated date
`status`	Jira status (stored on the index doc)
`rawText`	description + attachment text, packed to ~24k tokens (§4.2)

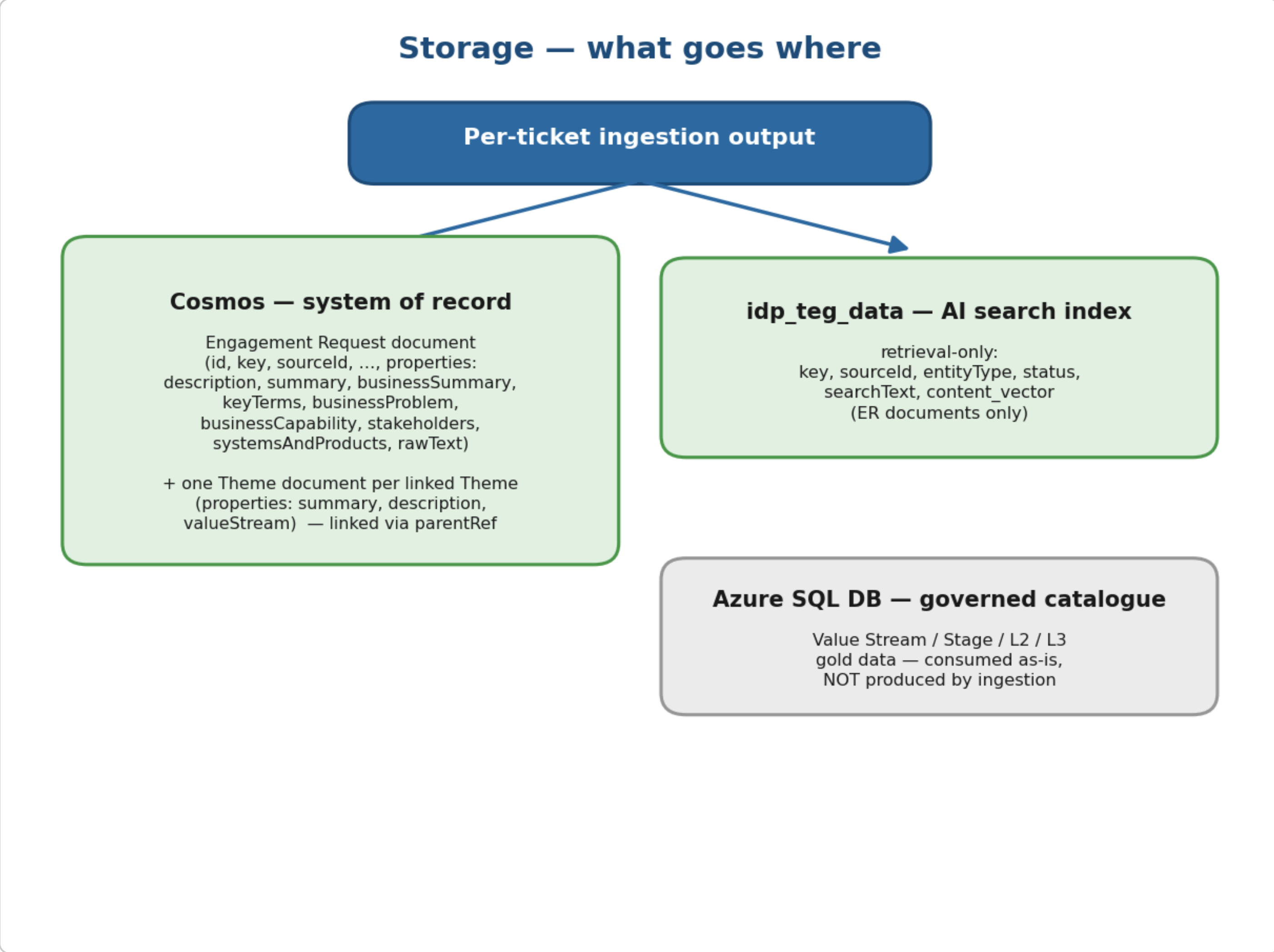
IDMT Engagement Request — from Condense (LLM)

field	source
`businessSummary`	LLM-generated business summary
`keyTerms`	domain terms & acronyms
`businessProblem`	business problem / pain point
`businessCapability`	desired capability / outcome
`stakeholders`	stakeholder groups
`systemsAndProducts`	referenced systems, platforms, products

Theme — from Jira

field	source
`key`	Jira issue key (GROUP-####)
`sourceId`	stable Jira internal id
`summary`	Theme title
`description`	Theme description
`valueStream`	the Theme's **Business Value Stream** field, formatted `"<name> {id}"`, taken **as-is** — no fuzzy matching, no LLM, no catalogue re-resolution
`creationDate`	Theme creation date
`insightsTime`	Theme last-updated date

5. Storage schema



5.1 Cosmos — Engagement Request document

field	description
`id`	document uuid
`key`	Jira issue key, e.g. `IDMT-####` (mutable business key)
`sourceId`	stable Jira internal id (e.g. 3364549); stable across IDMT-key changes
`source`	origin system — `JIRA` (uppercase)
`domain`	`WORKITEM` (uppercase)
`entityType`	`ENGAGEMENTREQUEST` (uppercase)
`createdAt` / `createdBy`	Cosmos creation date / actor (`createdBy` = `TEG-INGESTION`)
`lastModifiedAt` / `lastModifiedBy`	Cosmos modification date / actor (`lastModifiedBy` = `TEG-INGESTION`)
`parentRef`	the ER's own `sourceId` (an ER has no parent)
`properties`	nested object — extracted business context (below)

`properties` **:** `description` , `summary` , `creationDate` , `insightsTime` , `businessSummary` , `keyTerms` , `businessProblem` , `businessCapability` , `stakeholders` , `systemsAndProducts` , `rawText` .

5.2 Cosmos — Theme document

field	description
`id`	document uuid
`key`	Jira issue key (GROUP-####)
`sourceId`	stable Jira internal id
`source`	origin system — `JIRA` (uppercase)
`domain`	`WORKITEM` (uppercase)
`entityType`	`THEME` (uppercase)
`createdAt` / `createdBy`	Cosmos creation date / actor (`createdBy` = `TEG-INGESTION`)
`lastModifiedAt` / `lastModifiedBy`	Cosmos modification date / actor (`lastModifiedBy` = `TEG-INGESTION`)
`parentRef`	the parent IDMT ticket's `sourceId`
`properties`	nested object (below)

`properties` : `summary` (Theme title), `description` , `valueStream` (the linked Value Stream as the single `<name> {id}` string, e.g. `Resolve Appeal {VSR00074590}` — not a nested object), `creationDate` , `insightsTime` .

> Themes are **separate documents**, found via `parentRef` — not embedded as a `themes[]` array on the IDMT document.

5.3 AI Search index (idp_teg_data) — retrieval-only

Holds the historical Engagement-Request documents for retrieval.

field	description
`key`	Jira issue key, e.g. `IDMT-####` (mutable business key)
`sourceId`	stable Jira internal id (stable across IDMT-key changes)
`entityType`	`ENGAGEMENTREQUEST`
`status`	Jira status (Cancelled / To Do / In Progress …)
`searchText`	`businessSummary + businessProblem + businessCapability + keyTerms + stakeholders + systemsAndProducts`
`content_vector`	vectorized `searchText`

`content_vector` is produced by the `text-embedding-3-small-idp` embedding model (1536 dimensions). The index returns ranked ids; the full ticket details are read from Cosmos at query time.

5.4 Azure SQL DB — governed catalogue (consumed, not produced)

The Value Stream / Stage / L2 / L3 catalogue is the org's **gold data in Azure SQL**, read as-is at runtime. Ingestion does **not** define or store this catalogue.

6. Rules & conventions

- **Content is read directly from Jira** — the Theme's Value Stream is taken as-is from its Business Value Stream field; no fuzzy matching, no LLM, no catalogue re-resolution.
- **We store only the Theme's title, description, and Value Stream** — no Epics, Stages, L2/L3, or Business Needs are extracted or stored.
- **The index is retrieval-only** — `searchText` + `content_vector`; no labels or signals stored in it.
- **The Value Stream catalogue is Azure SQL gold data**, consumed as-is — not redefined or stored here.
- **Unit tests make no live Jira / Azure / LLM calls** — clients are injected (fakes).